

**DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING**

**SCHOOL OF COMPUTING**

**ABSTRACT SUBMISSION FORM**

**10214CS701 – MAJOR PROJECT**

**ACADEMIC YEAR: 2024-2025**

**SEMESTER: WINTER**

| Name of the Student                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | Student 1                                               | Student 2  | Student 3 |
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| VTU No                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | 20548                                                   | 19521      |           |
| Reg. No                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | 21UECS0073                                              | 21UECS0254 |           |
| Dept with Spl                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | CSE                                                     | CSE        |           |
| <b>TITLE OF THE PROJECT:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | <b>CERVICAL CANCER DETECTION USING MACHINE LEARNING</b> |            |           |
| <b>Name of the Supervisor : Dr.G.Prabakaran</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                                                         |            |           |
| <p><b>ABSTRACT :</b></p> <p>CERVICAL CANCER REMAINS A LEADING CAUSE OF MORTALITY AMONG WOMEN WORLDWIDE, AND ITS EARLY DETECTION IS CRITICAL FOR EFFECTIVE TREATMENT AND IMPROVED SURVIVAL RATES. THIS PROJECT LEVERAGES DEEP LEARNING TECHNIQUES, SPECIFICALLY CONVOLUTIONAL NEURAL NETWORKS (CNNs), FOR THE AUTOMATED DETECTION AND CLASSIFICATION OF CERVICAL CANCER USING THE PAP SMEAR DATASET. THE DATASET COMPRISES CYTOLOGICAL IMAGES CATEGORIZED INTO FOUR CLASSES: DYSKERATOTIC, KOILOCYTOTIC, METAPLASTIC, AND PARABASAL. THE IMAGES UNDERGO PREPROCESSING STEPS SUCH AS RESIZING, NORMALIZATION, AND AUGMENTATION TO ENHANCE THE MODEL'S ROBUSTNESS AND PERFORMANCE. THE CNN MODEL IS DESIGNED TO EXTRACT DETAILED FEATURES FROM THE PAP SMEAR IMAGES, ENABLING ACCURATE CLASSIFICATION. THE SYSTEM'S PERFORMANCE IS EVALUATED USING METRICS SUCH AS ACCURACY, PRECISION, RECALL, AND F1-SCORE. RESULTS INDICATE THE POTENTIAL OF THIS APPROACH IN AUTOMATING CERVICAL CANCER SCREENING, THEREBY REDUCING DIAGNOSTIC TIME AND AIDING PATHOLOGISTS IN MAKING PRECISE ASSESSMENTS. THIS STUDY HIGHLIGHTS THE SIGNIFICANCE OF INTEGRATING ARTIFICIAL INTELLIGENCE WITH MEDICAL DIAGNOSTICS TO IMPROVE HEALTHCARE OUTCOMES, PARTICULARLY IN RESOURCE-CONSTRAINED SETTINGS.</p> |                                                         |            |           |

STUDENT

PROJECT SUPERVISOR

PROJECT CO-ORDINATOR